CSC621 Data Structures and Algorithm

Lab Exercise 1 - Algorithmic Efficiency

Tutorial Questions

1. Evaluate the following code segments in terms of their algorithmic efficiency. Do not use Big O notation, but rather give a specific number for an answer in terms of time units. (Note: Assume that i++ count as 2 time units.)

```
x = 0;

for(i=0; i<3; i++)
x = x + i * 2;
```

2. Evaluate the following code segments in terms of their algorithmic efficiency. Do not give a specific number, but rather use Big O notation in terms of n.

```
i = 1;
while(i <= n/2) {
// some statements
i = i + 1;
}
```

3. Given the following expressions from analyzing algorithms, express them in Big O notation. (Hint: only the dominant term should appear within the parenthesis without any coefficients.)

   - log n + 2n + 3
   - n log n + n^2 + 3n
   - log n + n log n + n

Tutorial Questions

The Fibonacci Sequence is the series of numbers:

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

The next number is found by adding up the two numbers before it:

- the 2 is found by adding the two numbers before it (1+1),
- the 3 is found by adding the two numbers before it (1+2),
- the 5 is (2+3),
- and so on!

Ref: https://www.mathsisfun.com/numbers/fibonacci-sequence.html

Task: Write a recursive function using java to find the nth Fibonacci number.

Sample output:

The 0th fibonacci number is: 0

The 7th fibonacci number is: 13

The 12th fibonacci number is: 144